

MD. SHAHRIAR HOSSAIN JIHAD

Security-Focused Software Engineer | Application Security | Network Security | Malware Detection

Dhaka, Bangladesh | 0130-6891479 | devshahriarr@gmail.com

LinkedIn: linkedin.com/in/devshahriarr | GitHub: github.com/devshahriarr | Portfolio: devshahriar.com

PROFESSIONAL SUMMARY

Security-focused Software Engineer with 3.5+ years of professional experience in software engineering, backend systems, SaaS architecture, Linux production environments, and technical leadership, transitioning into a dedicated Cybersecurity career. Hands-on experience in web/API security testing, network traffic analysis, Linux server security investigation, malware hunting, vulnerability assessment, and secure software engineering using Wireshark, Nmap, Burp Suite, OWASP ZAP, SQLMap and Linux security tools.

Currently working on an AI-powered malware detection and security agent using the EMBER dataset, Python, Scikit-learn, XGBoost, LightGBM and FastAPI. Strong understanding of how real-world web applications, APIs, databases, and Linux infrastructure are designed, deployed and secured. Managed 100+ software projects, coordinated 20+ concurrent projects, led teams of up to 35 members, and contributed to approximately \$80K+ in revenue within 9 months.

CYBERSECURITY SKILLS

Application Security: OWASP Top 10, Web/API Security, Authentication & Authorization, Access Control, Input Validation, SQL Injection, Rate Limiting

Network Security: TCP/IP, Packet Analysis, Reconnaissance, Port Scanning, Service Enumeration

Tools: Wireshark, Nmap, Burp Suite, OWASP ZAP, SQLMap, Postman, Fail2ban

Linux Security: SSH Hardening, Log Analysis, Process/File Investigation, Cron/Service Analysis

Malware/AI Security: Malware Hunting, PE Analysis, EMBER, XGBoost, LightGBM, Prompt Injection, AI Threat Detection

Engineering & Delivery: Requirements Analysis, Feasibility Study, Solution Architecture, Database Design & ERD, Project Planning, Technical Documentation

Programming & Databases: Python, PHP, JavaScript, SQL, MySQL, PostgreSQL, Data Modeling

Frameworks & Infrastructure: FastAPI, Laravel, React, Next.js, Redis, Docker, Git, VPS

PROFESSIONAL EXPERIENCE

Programming: Python, PHP, JavaScript, MySQL, PostgreSQL, ERD, Data Modeling, SQL

Assistant Manager, Operations *July 2026 – Present*

Fire AI—Betopia Group (Previously Full-Stack Developer | October 2025 – June 2026)

- Promoted from Full-Stack Developer to Assistant Manager, Operations, currently coordinating a multidisciplinary technical team of approximately 20 members, with previous leadership responsibility for teams of up to 35 members.
- Have handled 100+ software projects across two organizations, currently overseeing 20+ concurrent projects and coordinating AI, backend, frontend, mobile, and UI/UX resources from requirements through delivery.
- Lead technical and project planning, requirements analysis, feasibility studies, solution architecture, database design and ERD development, development coordination, testing, deployment, and production troubleshooting.
- Conduct hands-on Linux server security investigations, including SSH authentication log analysis, suspicious process and file investigation, cryptomining-related malware hunting, access key review, and server hardening.
- Work directly with **250+ international clients**, handling requirements, technical consultation, project communication and delivery coordination in English; also produce PRDs, SRS documents, testing documentation, technical specifications and user guidelines
- Contributed to approximately \$80K+ in revenue generation within 9 months through technical leadership, project delivery, client communication, and operations management.

Lead Programmer & Project Manager *September 2023 – September 2025*

Eclipse IntelliTech

- Led development and delivery of commercial software across SaaS, ERP, POS, eCommerce, and business automation domains.
- Managed developers and coordinated multiple projects from requirements analysis and architecture through development, testing, deployment, and support.
- Designed and developed secure backend/API systems using Laravel/PHP, MySQL/PostgreSQL and Linux-based infrastructure.
- Performed application and API security testing covering authentication, authorization, access control, input validation, API behaviour and common web vulnerabilities.
- Contributed to handling 100+ software projects across two organizations, including projects personally developed and projects delivered through managed development teams.

Front-End Developer *January 2023 – August 2023*

Eclipse IntelliTech

- Developed full-stack web applications, REST APIs and database-driven business systems using Laravel/PHP, JavaScript and MySQL.
- Implemented authentication, authorization, input validation and access-control mechanisms while troubleshooting application and production issues.

HANDS-ON CYBERSECURITY PROJECTS

AI-Powered Malware Detection & Security Agent

Python | EMBER | Scikit-learn | XGBoost | LightGBM | FastAPI

Github Link: <https://github.com/devshahriar/AP-Powered-Malware-Hunting-project>

- Developing an AI/ML-based malware detection system for identifying malicious Windows PE files using the EMBER malware dataset.
- Working across feature processing, malware/benign classification, and model experimentation and evaluation using Scikit-learn, XGBoost and LightGBM.
- Designing a FastAPI inference service to analyse PE files and return Malware/Benign predictions with confidence, with the broader goal of AI-assisted automated threat detection.

Linux Server Malware Investigation & Hardening

Linux | Ubuntu | SSH | Fail2ban | System Logs

- Conducted hands-on investigation of suspicious activity and cryptomining-related malware on a Linux production server.
- Analysed SSH authentication logs, running processes, suspicious files, cron/system services and ownership/permission anomalies to investigate potential persistence and unauthorized activity.
- Performed SSH hardening, reviewed users and SSH keys, investigated brute-force attempts and validated Fail2ban-based protection.

Network & Web Security Testing

Wireshark | Nmap | Burp Suite | OWASP ZAP | SQLMap

- Performed hands-on packet capture and network traffic analysis using Wireshark to understand network and application-level communication.
- Used Nmap for network reconnaissance, port scanning and service enumeration in authorized environments.
- Conducted web/API security testing using Burp Suite and OWASP ZAP, including authentication, authorization, access-control and input-validation testing; used SQLMap for authorized SQL injection validation.

SECURITY LEARNING & PROFESSIONAL DEVELOPMENT

Cisco Networking Academy — Introduction to Cybersecurity *(In Progress)*

Focused on cybersecurity fundamentals, threats and vulnerabilities, defensive security concepts and security best practices.

EDUCATION

Diploma in Engineering, Computer Technology

Bogura Polytechnic Institute | 2022

B.Sc. in Computer Science & Engineering

Northern University Bangladesh | Expected 2026

LANGUAGES

Bangla: Native | **English:** Fluent | **Hindi:** Fluent